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MeSIN: Multilevel selective and interactive network for medication recommendation

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ABSTRACT

Recommending medications for patients using electronic health records (EHRs) is a crucial data mining task for an intelligent healthcare system. EHRs data, which comprise multiple temporal sequences (e.g., lab, diagnosis, and treatment sequences), encompass multilevel structure information, including flat structure information (e.g., sequential medical codes and their relationships) in each sequence and hierarchical structure information between sequences (e.g., information flow from lab sequence to diagnosis sequence or hierarchical codes relationships), which implicitly reflects a physician's decision process. However, existing studies focus on mining flat structure information, which is incomplete and hence biased. Therefore, to realize more accurate medication recommendations, this paper proposes a multilevel selective and interactive network (MeSIN) that exploits complete structure information of EHR data; in particular, at the core of our framework is an interactive long short-term memory network (InLSTM), which is developed to model the temporal dependencies in each flat-structured sequence via a recurrent mechanism and capture the hierarchical relationships of different sequences via a reinforced interactive network. Moreover, to reduce the impact of irrelevant information on the multilevel interaction process, an attentional selective module (ASM) is introduced to filter out noisy information and thereby focus on important sequence information at each timestamp. Finally, a global selective fusion module (GSFM) is proposed to infuse multisourced information embeddings into multilevel structural relationship-aware patient representation. Experimental results on a practical clinical dataset show that our framework consistently outperformed all baseline models in the medication recommendation task.

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1. Introduction

Medication-related mining tasks [1–3] have always been prevalent in both academia and industry. For instance, when patients visit a doctor, the doctor generally needs to diagnose and prescribe medicines according to the electronic health records (EHRs) of the patients. However, owing to the patient-specific characteristics of EHR data, one-on-one diagnosis and treatment would become overwhelming for inexperienced (or experienced) doctors. Therefore, automatic medication recommendation would be an effective solution. In particular, deep learning [4] techniques enable the development of such an intelligent system to assist doctors in making effective and safe medication prescriptions for certain patients, as shown in Fig. 1(c). However, the complexity EHRs data renders a challenging medication recommendation task.

As illustrated in Fig. 1(b), EHRs data generally comprise multiple temporal and heterogeneous sequences including but not limited to lab sequence, diagnosis sequence, and treatment sequence, which encompass not only the flat structure information such as the sequential medical codes and their relationships in each sequence, but also the hierarchical structure information between sequences that implicitly reflects a physician's decision process. Laboratory results can provide sufficient hints for certain diseases, for example, when the anion gap is abnormally high, creatinine is abnormally low, and aspartate is abnormally high, it demonstrates the presence of pulmonary diseases, such as respiratory infections and pulmonary emphysema. Correspondingly, the patient needs treatments such as glucose, sodium bicarbonate, xylitol, and budesonide. This example indicates the inherent hierarchical relationships between heterogeneous medical sequences. Although recent studies [5] have testified the importance of such multilevel structures, most existing medication recommendation studies [6–8] put more effort into mining the former flat-structured information to model the mapping

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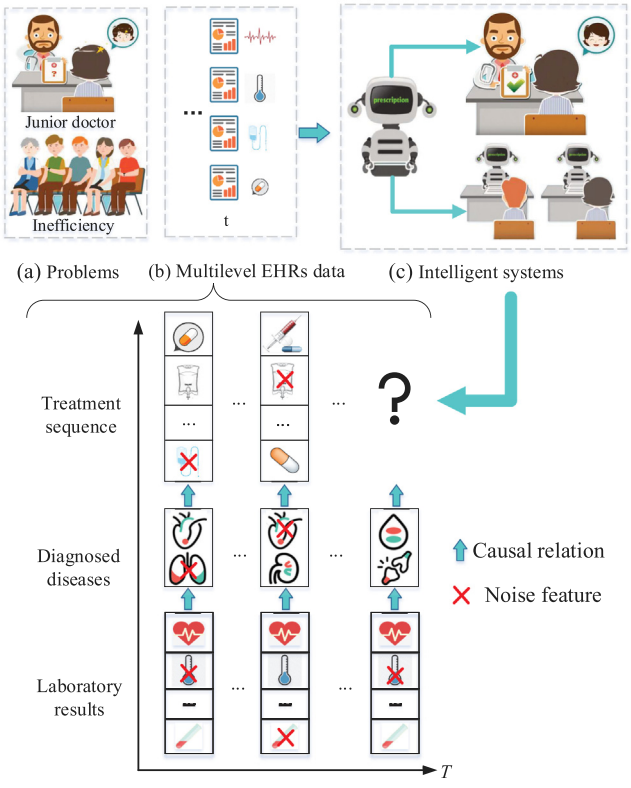


Fig. 1. Urgent need for developing an intelligent system with multilevel EHRs data and corresponding complexity.

relations between input medical information and recommended medications, but overlooked the latter paramount hierarchically correlative structure information between sequences. Thus, apart from modeling the temporal dependencies in each flat-structured sequence, concurrently capturing the hierarchical relationships between multiple sequences has become a critical challenge for modeling an intelligent medication recommendation task.

However, it is nontrivial to capture the hierarchical relationships between multilevel sequences except for respectively modeling their own temporal dependencies. Existing methods employ either the attention mechanism [9,10] implicitly or a specially designed interaction structure [11,12] to capture the correlations of multilevel sequences explicitly. Note that, while explicit interaction structures in DE-LSTM [11] and RAHM [12] are constructed, they only consider the effects of related input sequences on the memory cell state while neglecting the influence on the current input state, which belongs to an incompletely interactive process. The above-mentioned approaches neglect to model the inherent hierarchical multilevel structure. Thus, in this study, the sequential interactions between sequences are further reinforced through specifically designed interactive networks that can simultaneously consider the historical relevance and current correlations in the temporal sequence learning network.

In addition, there are unavoidable noisy information existing in the multisourced heterogeneous EHRs data, which have an unfavorable impact on the subsequent hierarchically sequential interaction process. Thus, reducing the adverse influence of irrelevant information is another critical challenge that could inhibit medication recommendation performance. However, few deep learning studies in health informatics incorporate feature selections into the learning process, except for LSAN [13], which assigns flexible attention weights to different diagnosis codes based on their relevance to corresponding diseases. However, doctors pay more attention to only a few critical factors and neglect irrelevant medical indicators or historical medical codes. In other words, irrelevant features should be deleted in the decision-making process, if possible, and unimportant historical medical codes should be paid less attention. Thus, the general attention mechanism might not be appropriate for the learning process.

1.1. Contributions

To tackle the aforementioned challenges, in this study, a multilevel selective and interactive network, called MeSIN, is proposed for medication recommendation. In particular, the framework can concurrently capture the temporal correlations within each flat-structured sequence via the recurrent mechanism and hierarchical relationships between sequences via a novel reinforced interactive network. Moreover, to reduce the adverse influence of irrelevant input information on the multilevel interaction process, the selective module is introduced into every sequential learning network in the framework to filter out possible noisy information by automatic and intelligent selections. Then, the learned multisourced information embeddings are infused into the patient representation through a globally selective fusion module. Finally, the comprehensive patient representation obtained could be leveraged for medication recommendations. Thus, the main contributions in this study are as follows:

- The study is the first to solve the medications recommendation task via a hierarchically multilevel learning framework, MeSIN, which fully leverages the inherent multilevel structure of EHR data to learn a comprehensive patient representation for reasonable medication recommendation. It imitates the physician's decision process to some extent.
- An interactive long-short term memory network (InLSTM) is proposed to effectively reinforce the interactions of temporal and heterogeneous sequences with the help of recurrent neural structure, a new calibrated memory-augmented cell and an enhanced input gate.
- An improved attentional selective module (ASM) is incorporated to reduce the adverse impact of irrelevant information, and thus to enhance the real influence of important input information on the interaction process in InLSTM.
- A self-attention based global selective fusion module (GSFM) is designed to integrate the obtained heterogeneous embeddings into patient representation according to respective importance and minimize the adverse effects induced by the irrelevant information.

2. Related works

Related studies in healthcare informatics are reviewed from the following three perspectives: medication recommendation, attention mechanism in health informatics, and sequence modeling in health informatics.

2.1. Medication recommendation

Researchers have been trying to resort to advanced information technologies to tackle the various healthcare-related problems [14,15]. For example, artificial intelligence, particularly deep learning-related techniques, has been successfully applied in various aspects of the medical domain [16], such as the medication recommendation application. Leveraging recommender algorithms [17–19] to recommend rational and effective medications in time for patients, as a paramount recommendation task in the

health domain, has been widely researched. Shabbir et al. [20] developed a smart medication recommendation model using association rules for data mining. Shang et al. [6] categorized medication recommendation-related tasks into instance-based and longitudinal sequential recommendation methods. Instance-based methods are based on current disease progression in patients. For example, Zhang et al. [21] formulated a medication recommendation task as a sequential decision-making problem and leveraged a recurrent decoder to model label dependency. Wang et al. [8] addressed the recommendation issues by casting the task as an order-free Markov decision process (MDP) problem. However, they all ignored valuable historical information. Until now, longitudinal sequential recommendation methods have mainly considered the impact of historical medical records by modeling their temporal dependencies. For instance, Jin et al. [22] developed three different LSTMs to model heterogeneous data interactions to predict next-period prescriptions. Shang et al. [6] incorporated historical diseases and procedure codes, as well as medication records, in their model. Shang et al. [23] considered hierarchical knowledge of diagnoses and medications to enhance the code representation for medication recommendations. An et al. [12] formulated a medication prediction task as a hierarchical multitask learning framework to improve the interpretability of the predicted results. However, few of them simultaneously consider all multiple heterogeneous sequences and the correlations between them in the decision-making of medication recommendations.

2.2. Attention mechanism in health informatics

An attention mechanism has been proposed to automatically assign importance scores according to information relevance. In this case, larger weights indicate that the corresponding vectors are more relevant for generating the output. Owing to its powerful ability, the attention mechanism has been widely used in various neural network-based applications such as language understanding tasks [24,25] and computer vision problems [26,27]. Similarly, attention mechanisms in health informatics have been prevalent in predictive modeling. For instance, GRAM [28], KAME [29], and G-BERT [23] leveraged the attention mechanism to integrate domain knowledge into disease or medication code representations for better performance. Retain [30], Dipole [31], Timeline [32], and LSAN [13] introduced an attention mechanism to model disease progression by considering the dependencies among visits and provide some interpretable insights. In addition, GCT [5] was equipped with advanced attention networks, that is, Transformer [33], to build the correlations between medical codes from every visit based on the learned attention weights. Likewise, AMANet [10] utilized multiple attention networks, including self-attention and inter-attention, to capture the intra-view interaction and inter-view interaction. However, the attention mechanism used in the above models generated dense attention weights without zero weight values, which means that they cannot filter out the noise information and focus on the critical aspects.

2.3. Sequence modeling in health informatics

Owing to the complexity of the clinical scenarios, as shown in Fig. 1, EHR systems in hospitals accumulate complex temporal and heterogeneous sequences. Existing studies in health informatics have widely utilized the temporal sequential records from EHRs to solve healthcare problems such as predicting disease progression [13,34–36], medication recommendations [6,11,21], and clinical trial recruitment [37,38]. However, most of the studies, such as T-LSTM [39], MNN [35], and LSAN [13] focused on modeling the temporal dependencies of multiple visits of homogeneous sequences such as the history disease sequence. While the medication recommendation task involves multiple temporal and heterogeneous sequences, not only the temporal intra-dependencies but also the inter-correlations between the sequences should be considered when modeling the sequence learning process. Although GAMENet [6] utilized two medical sequences to model the temporal dependencies for medication recommendations, it did not consider the correlations of sequences. DMNC [9] presented a two-view sequential learning model to model complex interactions. However, the complex differentiable neural computer (DNC) blocks used in DMNC [9] do not explicitly model sequential interactions. In contrast, Jin et al. [11] developed three heterogeneous LSTM models to model the correlations of different types of medical sequences by connecting hidden neurons, but neglect the impact on patient's current status. MiME [34] modeled the inherent multilevel structures of medical codes by incorporating the relationships between the diagnoses and their treatments into patient visit representations. AMANet [10] utilized multiple attention networks to capture the intra- and inter-view interactions of heterogeneous and temporal sequences, but overlooked the multilevel nature of EHR data.

3. Methods

3.1. Problem definitions

To facilitate the latter introduction of our computational methods and generalize the applicable dataset, we define the data from the electronic health record system (EHR) using the mathematical symbols as follows: The longitudinal EHR data contain a large number of patient records, and each patient can be represented as a sequence of multivariate observations: $\mathcal{P} = \left\{\mathcal{X}^1, \mathcal{X}^2, \ldots, \mathcal{X}^{t_n}\right\}$, where $n \in \{1, 2, \ldots, N\}$, N is the total number of patients, and t_n is the number of visits for the nth patient. Without loss of generality, we describe the model for a patient, and the subscript (n) will be dropped whenever it is unambiguous. Each visit $\mathcal{X}^t$ consists of sequential laboratory indicator-wise results $\mathcal{L}^t = \left\{[l_{11}, \ldots, l_{1T}], \ldots, [l_{q1}, \ldots, l_{qT}]\right\}$, where l_{qT} denotes the qth indicator result at Tth timestamp within tth visit, and categorized data including $\mathcal{C}_d^t \subset \mathcal{C}_d$ (a union set of diagnoses codes) and $\mathcal{C}_m^t \subset \mathcal{C}_m$ (a union set of medications codes). For simplicity, we use $\mathcal{C}_*^t$ to represent a unified definition of medical codes. $\mathcal{C}_*$ denotes the medical code set, and $|\mathcal{C}_*|$ denotes the size of the medical code set. $\mathbf{c}_*^j$ is the jth medical code in $\mathcal{C}_*$.

Problem Definition 1 (*Medication Recommendation*)**.** Given the historical visit records of a patient $\left\{\mathcal{X}^1, \mathcal{X}^2, \ldots, \mathcal{X}^{t-1}\right\}$, the current laboratory results $\mathcal{L}^t$, and diagnosed diseases $\mathcal{C}_d^t$, our goal is to recommend reasonable medications by generating the multilabel output $\hat{\boldsymbol{y}}_t^m \in \{0, 1\}^{|\mathcal{C}_m|}$.

$$\hat{\boldsymbol{y}}_t^m = f(\left\{\mathcal{X}^1, \mathcal{X}^2, \ldots, \mathcal{X}^{t-1}\right\}, \mathcal{L}^t, \mathcal{C}_d^t). \tag{1}$$

3.2. Multilevel selective and interactive network

We propose a novel architecture, MeSIN, for implementing the medication recommendation task. As shown in Fig. 2, MeSIN is a multilevel learning framework, which comprises two steps. In step **I**, the hierarchical historical information embedding learning process begins with the laboratory results embedding the learning module, followed by the diagnosis code embedding module, and then the medication code embedding module. In step **II**, the global selective fusion module is utilized to infuse the learned heterogeneous embeddings into the patient representation according to the selected weights.

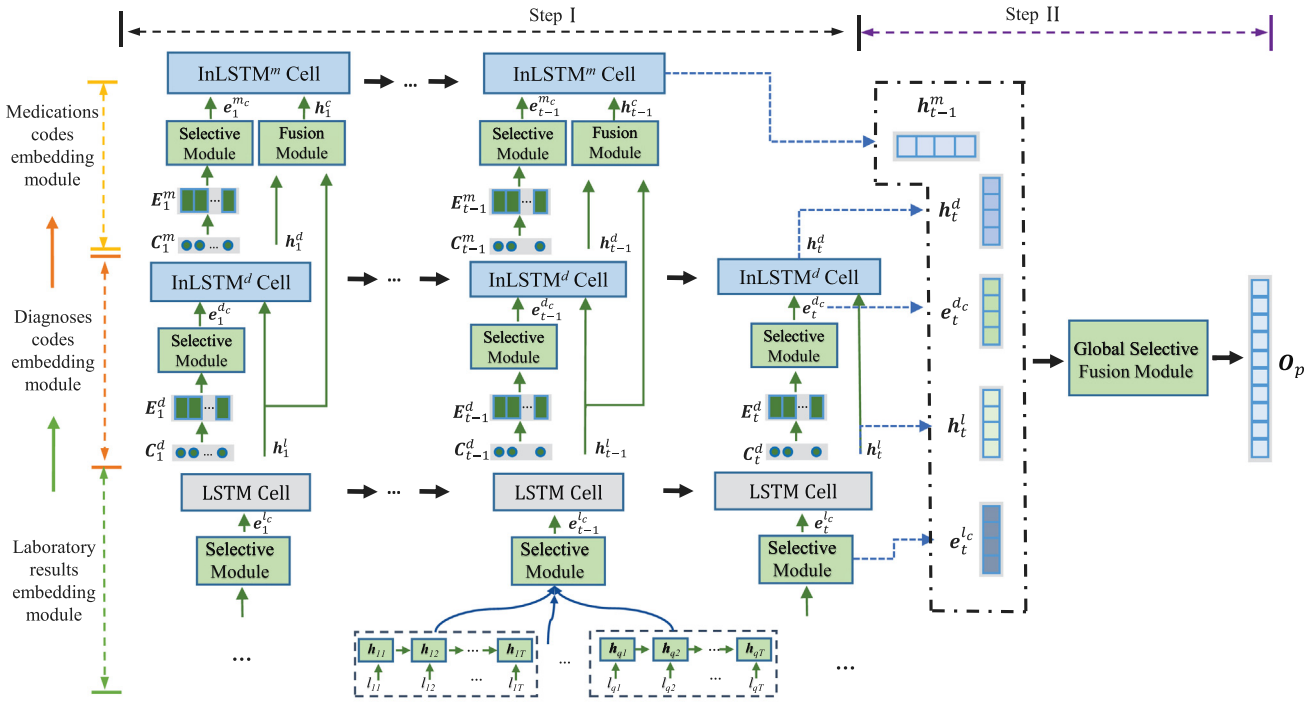


Fig. 2. Architecture of MeSIN. Overall, from the bottom up, MeSIN comprises three hierarchically correlative modules and a global fusion module. The laboratory results embedding module first projects the temporal sequence of each laboratory indicator $\{l_{q1}, \ldots, l_{qT}\}$ into embedded vectors $\{\boldsymbol{h}^l_{1T}, \ldots, \boldsymbol{h}^l_{qT}\}$ via a multichannel GRU, and then uses the attentional selective module to compute the enhanced embedding $\boldsymbol{e}^{l_c}_t$ which would be input into LSTM to obtain the visit-level embedding $\boldsymbol{h}^l_t$; The diagnoses codes embedding module and medications codes embedding module respectively contain three substructures: the medical codes embedding layer for mapping the medical codes set $\boldsymbol{C}^t_*$ into dense embeddings set $\boldsymbol{E}^t_*$, the attentional selective module for computing the enhanced embedding $\boldsymbol{e}^{*_c}_t$, and the interactive LSTM (InLSTM) for calculating the visit-level embedding $\boldsymbol{h}^*_t$. Finally, the global selective fusion module is used to infuse the learned multisourced embeddings into patient representation $\boldsymbol{O}_p$ for medication recommendation.

3.2.1. Laboratory sequence embedding module

As shown in Fig. 2, the module mainly consists of three key parts: a multichannel time-series embedding layer, an attentional selective layer, and a temporal sequence learning network.

Multichannel time-series embedding layer. As the meanings of particular clinical features for patients in diverse medical conditions are different, the progression of laboratory indicators is distinct. Thus, the embedding of sequential features representing the indicator change progress is distinct from each other. Here, inspired by ConCare [40], to capture the specified progression for every laboratory indicator, we employ the multichannel time-series embedding layer to embed the sequence of each laboratory indicator feature separately by multichannel GRUs:

$$\boldsymbol{h}^l_{qT} = \mathrm{GRU}_q\left(l_{q1}, l_{q2}, \ldots, l_{qT}\right), \tag{2}$$

where $\{l_{q1}, l_{q2}, \ldots, l_{qT}\}$ denote the time series and $\boldsymbol{h}^l_{qT}$ represents the embedded vectors of feature q. Therefore, all the embedded vectors of the time series of indicator features $\{\boldsymbol{h}^l_{1T}, \ldots, \boldsymbol{h}^l_{qT}\}$ can be acquired in the same manner. To reduce clutter, the superscript (t) representing the results generated at the tth visit will be dropped whenever it is unambiguous.

ASM for laboratory results embeddings selection. For each sequence of the laboratory results, we obtain the corresponding embeddings $\boldsymbol{h}^l_{1T}, \ldots, \boldsymbol{h}^l_{qT}$. However, in clinical scenarios, doctors pay attention to only a few indicators according to clinical experience, which can effectively improve work efficiency.

In this case, the attention mechanism that computes the attention weights using the **softmax function** [41] might be inappropriate because it results in dense attention alignments that are wasteful and makes models less interpretive. Therefore, we introduce sparse attention using **entmax** [42] to compute attention weights for ASM of MeSIN to increase focus on relevant source medical code embeddings and make the model more interpretable. Here, we employ the specially proposed ASM to compute the enhanced laboratory results embedding $\boldsymbol{e}^{l_c}_t$:

$$\begin{aligned}
\boldsymbol{e}^{l_c}_t &= \sum_{i=1}^{q} \alpha^l_i \boldsymbol{h}^l_{iT} (\alpha^l_i \in \boldsymbol{\alpha^l}), \\
\boldsymbol{\alpha^l} &= \alpha\text{-entmax}([\alpha^l_1, \alpha^l_2, \ldots, \alpha^l_q], \gamma_l), \\
\alpha^l_i &= \tanh(\boldsymbol{W}_{l_a}{}^{\top} \boldsymbol{h}^l_{iT} + \boldsymbol{b}_{l_a}),
\end{aligned} \tag{3}$$

where $\boldsymbol{W}_{l_a} \in \mathbb{R}^d$ and $\boldsymbol{b}_{l_a} \in \mathbb{R}$ are the parameters of the ASM to be learned. $\alpha - entmax$ [42] is a special method of entmax that allows us to find the optimal equilibrium point by controlling the value of γ_l. For $\gamma_l > 1$, as the value increases, entmax tends to produce sparse probability distributions, yielding a function family interpolating between softmax and sparsemax. In this manner, we can compute all the enhanced laboratory results embeddings $\{\boldsymbol{e}^{l_c}_1, \ldots, \boldsymbol{e}^{l_c}_t\}$ at each timestamp.

Temporal sequence learning network. To further capture the temporal dependency of multivisit laboratory results, the enhanced laboratory results embeddings $\{\boldsymbol{e}^{l_c}_1, \ldots, \boldsymbol{e}^{l_c}_t\}$ are input into the temporal sequence learning network to combine with the historical laboratory results:

$$\boldsymbol{h}^l_t = \mathrm{LSTM}_L\left(\boldsymbol{h}^l_{t-1}, \boldsymbol{e}^{l_c}_t\right), \tag{4}$$

where LSTM_L represents the long-short temporal neural network (LSTM) for capturing the temporal dependency of laboratory examination sequence, $\boldsymbol{h}^l_t$ denotes the obtained visit-level laboratory results embedding containing the history information at tth visit. Using the same calculations for the remaining timestamps, we can finally obtain all the historical laboratory results

embeddings $\{\boldsymbol{h}_1^l, \ldots, \boldsymbol{h}_t^l\}$, which will be input into the following embedding modules.

3.2.2. Diagnoses codes embedding module

After checking the laboratory results, the doctors tended to retrieve the historically diagnosed diseases and combine them with the current disease condition for comprehensive decision-making. Similarly, as shown in Fig. 2, we design a module that contains three critical parts: a diagnosis code embedding layer, an attentional selective module, and a novel temporal sequence interactive learning network.

Diagnosis code embedding layer. Taking the timestamp t as an example, MeSIN first encodes each diagnosis code $\boldsymbol{c}_i^d$ into a dense representation vector $\boldsymbol{e}_i^d \in \mathbb{R}^d$ as

$$\boldsymbol{e}_i^d = \boldsymbol{W}_e^d \boldsymbol{c}_i^d, \tag{5}$$

where $\boldsymbol{W}_e^d \in \mathbb{R}^{d \times |\mathcal{C}_*|}$ is the embedding matrix of medical codes that needs to be learned, d is the size of the embedding dimension, and $|\mathcal{C}_*|$ is the size of the medical code set. Thus, for the diagnosis code set $\mathcal{C}_d$, it can be represented by a collection of dense representation vectors $\boldsymbol{E}^d = [\boldsymbol{e}_1^d, \ldots, \boldsymbol{e}_{|\mathcal{C}_d|}^d] \in \mathbb{R}^{d \times |\mathcal{C}_d|}$. Then, for the ith visit, we can obtain a dense embedding set $\boldsymbol{E}_t^d = [\boldsymbol{e}_1^d, \ldots, \boldsymbol{e}_m^d] \in \mathbb{R}^{d \times |m}$, in which each embedding vector is extracted from $\boldsymbol{E}^d$ if it exists in the ith visit.

ASM for diagnoses codes embeddings selection. However, as discussed before, not every historical disease has an impact on the future disease risk, we should assign different relevance scores to each code embedding according to their degree of importance. Here, we also leverage ASM for the diagnosis code embedding selection. The enhanced diagnosis code embedding $\boldsymbol{e}_t^{d_c}$ can be calculated as

$$\begin{aligned}
\boldsymbol{e}_t^{d_c} &= \sum_{i=1}^{m} \alpha_i^d \boldsymbol{e}_i^d (\alpha_i^d \in \boldsymbol{\alpha}^{\boldsymbol{d}}), \\
\boldsymbol{\alpha}^{\boldsymbol{d}} &= \alpha\text{-entmax}([\alpha_1^d, \alpha_2^d, \ldots, \alpha_m^d], \gamma_d), \\
\alpha_i^d &= \tanh(\boldsymbol{W}_{d_a}{}^{\top} \boldsymbol{e}_i^d + \boldsymbol{b}_{d_a}),
\end{aligned} \tag{6}$$

where $\boldsymbol{W}_{d_a} \in \mathbb{R}^d$, and $\boldsymbol{b}_{d_a} \in \mathbb{R}$ are the parameters of the ASM to be learned. γ_d is the hyperparameter of $\alpha - entmax$ [42] in this module. In this manner, we can compute all the enhanced diagnosis code embeddings $\boldsymbol{E}^{d_c} = \{\boldsymbol{e}_1^{d_c}, \ldots, \boldsymbol{e}_t^{d_c}\}$.

InLSTM in diagnoses codes sequence learning. In addition to modeling the temporal dependency of a single sequence, we should also consider the interactions of multiple sequences in the sequence learning network. As discussed before, the laboratory results could be regarded as critical references when making diagnoses by doctors. Hence, the sequence of laboratory results should be used to control the diagnosed disease sequence learning process. Therefore, this type of network will adopt two input sequences: one is the primary input $\boldsymbol{x}_t^p$ of the sequence learning network, such as the diagnosis code embedding $\boldsymbol{e}_t^{d_c}$, and the other is the auxiliary input $\boldsymbol{x}_t^a$ to assist in controlling the primary sequence learning process, such as the learned visit-level laboratory results embedding $\boldsymbol{h}_t^l$.

Therefore, the basic LSTM model [43] is not appropriate under such circumstances. Inspired by LSTM-DE [22], we propose a novel interactive LSTM network (**InLSTM**), as shown in Fig. 3, to reinforce the interaction process of two associated sequences, which brings in two novel components including a calibrated memory-augmented cell and an enhanced input gate. It can be defined as follows:

$$\mathbf{h}_t = \text{InLSTM}\left(\mathbf{h}_{t-1}, \mathbf{x}_t^p, \mathbf{x}_t^a\right), \tag{7}$$

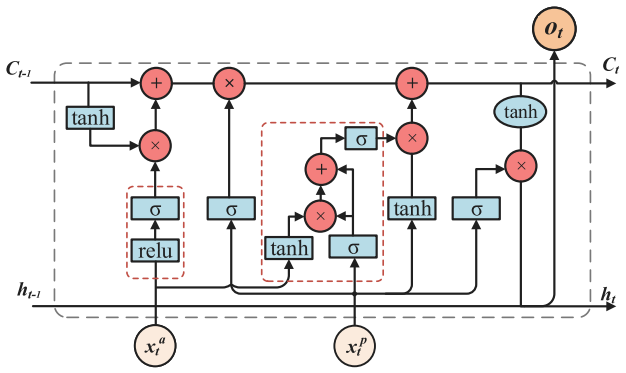


Fig. 3. Structure of InLSTM.

where the detailed mathematical expression of InLSTM is:

$$\begin{aligned}
\begin{bmatrix} \tilde{\boldsymbol{C}}_t \\ \boldsymbol{o}_t \\ \boldsymbol{i}_t \\ \boldsymbol{f}_t \end{bmatrix} &= \begin{bmatrix} \tanh \\ \sigma \\ \sigma \\ \sigma \end{bmatrix} \left(\boldsymbol{W} \begin{bmatrix} \boldsymbol{x}_t^p \\ \boldsymbol{h}_{t-1} \end{bmatrix} + \boldsymbol{b} \right), \\
\boldsymbol{C}_t &= \boldsymbol{f}_t * (\boldsymbol{C}_{t-1} + \tilde{\boldsymbol{C}}_t^e) + \boldsymbol{i}_t^e * \tilde{\boldsymbol{C}}_t, \\
\boldsymbol{h}_t &= \boldsymbol{o}_t * \tanh(\boldsymbol{C}_t),
\end{aligned} \tag{8}$$

where $\tilde{\boldsymbol{C}}_t^e$ denotes the calibrated memory-augmented cell state, calculated using Eq. (9), and $\boldsymbol{i}_t^e$ represents the enhanced input gate, which can be computed using Eq. (10).

$$\begin{aligned}
\boldsymbol{d}_t &= \tanh(\boldsymbol{W}_{enh} \boldsymbol{C}_{t-1} + \boldsymbol{b}_{enh}), \\
\boldsymbol{d}_t^e &= \sigma(\boldsymbol{U}_d^r)\, \text{ReLU}(\boldsymbol{W}_d^r \boldsymbol{x}_t^a), \\
\tilde{\boldsymbol{C}}_t^e &= \boldsymbol{d}_t * \boldsymbol{d}_t^e,
\end{aligned} \tag{9}$$

$$\begin{aligned}
\hat{\boldsymbol{i}}_t^e &= \tanh(\boldsymbol{W}_i^e \boldsymbol{x}_t^a + \boldsymbol{b}_i^e), \\
\tilde{\boldsymbol{i}}_t^e &= \boldsymbol{i}_t * \hat{\boldsymbol{i}}_t^e, \\
\boldsymbol{i}_t^e &= \sigma(\tilde{\boldsymbol{i}}_t^e + \boldsymbol{i}_t) \\
&= \sigma((\boldsymbol{i}_t^e + 1)\boldsymbol{i}_t),
\end{aligned} \tag{10}$$

where $\boldsymbol{d}_t$ indicates the obtained history memory state, and $\boldsymbol{d}_t^e$ denotes the calibrated gate calculated with auxiliary input $\boldsymbol{x}_t^a$. Subsequently, $\boldsymbol{d}_t^e$ is used to obtain the calibrated memory-augmented cell state value $\tilde{\boldsymbol{C}}_t^e$ by multiplying $\boldsymbol{d}_t$. In this manner, the calibrated gate can selectively assign more weights to the representative and predictive memory neurons while suppressing the unimportant neurons. In addition to the primary input $\boldsymbol{x}_t^p$ itself, it is also influenced by the auxiliary input $\boldsymbol{x}_t^a$. Under such circumstances, the calibrated auxiliary input $\hat{\boldsymbol{i}}_t^e$ is introduced to calculate the auxiliary influence score $\tilde{\boldsymbol{i}}_t^e$ by multiplying the normal input gate $\boldsymbol{i}_t$. Finally, the enhanced input gate $\boldsymbol{i}_t^e$ is computed by adding the auxiliary influence score $\tilde{\boldsymbol{i}}_t^e$ and the normal input gate $\boldsymbol{i}_t$, and then adjusted to a value between 0 and 1 via a sigmoid function.

Therefore, for the diagnosis code embedding module, we can calculate the final visit-level diagnoses codes embedding $\boldsymbol{h}_t^d$ by fusing with the historically diagnosed disease as

$$\boldsymbol{h}_t^d = \text{InLSTM}_d\left(\boldsymbol{h}_{t-1}^d, \boldsymbol{e}_t^{d_c}, \boldsymbol{h}_t^l\right), \tag{11}$$

where InLSTM_d denotes the proposed InLSTM (Eq. (7)) in this module, which is used to capture the correlations between the primary input of the diagnosis code embedding sequence and the auxiliary input of the laboratory results embedding sequence.

3.2.3. Medications codes embedding module

Similar to the previous hierarchy, the diagnosis code embedding module, the medication code embedding module still comprises three main parts: a code embedding layer, an attentional selective module, and a temporal sequence interactive learning network.

Medication code embedding layer. In this module, MeSIN first encodes each medication code $\boldsymbol{c}_z^m$ into a dense embedding vector $\boldsymbol{e}_z^m \in \mathbb{R}^d$ as

$$\boldsymbol{e}_z^m = \boldsymbol{W}_e^m \boldsymbol{c}_z^m, \tag{12}$$

where $\boldsymbol{W}_e^m \in \mathbb{R}^{|\mathcal{C}_*| \times d}$ is the embedding matrix of the medical codes that must be learned. Then, we can obtain the dense medication code embedding set $\boldsymbol{E}_z^m = [\boldsymbol{e}_1^m, \ldots, \boldsymbol{e}_n^m] \in \mathbb{R}^{d \times |n}$, in which each embedding is extracted from $\boldsymbol{E}^m = [\boldsymbol{e}_1^m, \ldots, \boldsymbol{e}_{|\mathcal{C}_m|}^m] \in \mathbb{R}^{|\mathcal{C}_m| \times d}$ if it exists in the ith visit.

ASM for medications codes embeddings selection. As mentioned before, MeSIN needs to filter out the noise coming from irrelevant historical medication code sets at each timestamp. In this case, we assign different relevance scores on different code embeddings using the attentional selective module for computing the enhanced medications set embedding $\boldsymbol{e}_{t-1}^{m_c}$:

$$\begin{aligned} \boldsymbol{e}_{t-1}^{m_c} &= \sum_{i=1}^{n} \alpha_i^m \boldsymbol{e}_i^m (\alpha_i^d \in \boldsymbol{\alpha^d}), \\ \boldsymbol{\alpha^m} &= \alpha\text{–entmax}([\alpha_1^m, \alpha_2^m, \ldots, \alpha_n^m], \gamma_m), \\ \alpha_i^m &= \tanh(\boldsymbol{W}_{m_a}{}^{\top} \boldsymbol{e}_i^m + \boldsymbol{b}_{m_a}), \end{aligned} \tag{13}$$

where $\boldsymbol{W}_{m_a} \in \mathbb{R}^d$, and $\boldsymbol{b}_{m_a} \in \mathbb{R}$ are the parameters of the ASM to be learned. γ_m is the hyperparameter of $\alpha - entmax$ [42] in this module. Likewise, we can compute the enhanced medication code embedding sequence $\boldsymbol{E}^{m_c} = \{\boldsymbol{e}_1^{m_c}, \ldots, \boldsymbol{e}_{t-1}^{m_c}\}$ at historical timestamps.

InLSTM in medications codes sequence learning. Finally, to capture the temporal dependency of historical medications, the enhanced medication code embedding sequence $\boldsymbol{E}^{m_c}$ is treated as the primary input of the sequence learning network. In addition, recommending medications is essentially a comprehensive decision-making process, and the historically prescribed medications must be affected by laboratory results and diagnosed diseases. Therefore, the sequences of laboratory results and diagnosed diseases were used as auxiliary inputs to control the sequence learning process. Then, we can calculate the final visit-level medication code embedding $\boldsymbol{h}_{t-1}^m$ by fusing the historical medications using the InLSTM Eq. (7), as follows:

$$\boldsymbol{h}_{t-1}^m = \text{InLSTM}_m \left(\boldsymbol{h}_{t-2}^m, \boldsymbol{e}_{t-1}^{m_c}, \boldsymbol{h}_{t-1}^c\right), \tag{14}$$

where InLSTM_m denotes the proposed InLSTM in this module, which is used to capture the correlations between the primary input of the medication code embedding and the auxiliary input $\boldsymbol{h}_{t-1}^c$. Further, the auxiliary input $\boldsymbol{h}_{t-1}^c$ is calculated using a fusion module:

$$\boldsymbol{h}_{t-1}^c = \sigma\left(\boldsymbol{W}_c^d \boldsymbol{h}_{t-1}^d + \boldsymbol{W}_c^l \boldsymbol{h}_{t-1}^l\right), \tag{15}$$

where $\boldsymbol{W}_c^d, \boldsymbol{W}_c^l \in \mathbb{R}^{d \times d}$, σ denotes the activation function *tanh*. $\boldsymbol{h}_{t-1}^l$ and $\boldsymbol{h}_{t-1}^d$, respectively, represent the obtained visit-level laboratory results embedding using Eq. (4), and diagnosed disease embedding using Eq. (11).

3.2.4. Global selective fusion module

In step **I**, by modeling the hierarchically interactive temporal sequence learning process, we can obtain the visit-level laboratory results embedding $\boldsymbol{h}_t^l$, diagnosis codes embedding $\boldsymbol{h}_t^d$, and the medication codes embedding $\boldsymbol{h}_{t-1}^m$. All of the above three types of embeddings have incorporated corresponding historical information. To recommend medications at the current timestamp, the current enhanced laboratory results embedding $\boldsymbol{e}_t^{l_c}$ and diagnosis code embedding $\boldsymbol{e}_t^{d_c}$ should be given more attention when making final decisions.

To effectively fuse the above five heterogeneous embeddings according to their importance scores and minimize the effect introduced by irrelevant information as much as possible, in step **II**, we design a global selective fusion module, which is realized by a self-attention mechanism. Because the five types of embeddings are heterogeneous, we first calculate the information importance scores by themselves as

$$\begin{bmatrix} \alpha_m \\ \alpha_d \\ \alpha_l \\ \alpha_{dc} \\ \alpha_{lc} \end{bmatrix} = \begin{bmatrix} \boldsymbol{W}_m^a \\ \boldsymbol{W}_d^a \\ \boldsymbol{W}_l^a \\ \boldsymbol{W}_{dc}^a \\ \boldsymbol{W}_{lc}^a \end{bmatrix} * \begin{bmatrix} \boldsymbol{h}_{t-1}^m \\ \boldsymbol{h}_t^d \\ \boldsymbol{h}_t^l \\ \boldsymbol{e}_t^{d_c} \\ \boldsymbol{e}_t^{l_c} \end{bmatrix} + \begin{bmatrix} \boldsymbol{b}_m^a \\ \boldsymbol{b}_d^a \\ \boldsymbol{b}_l^a \\ \boldsymbol{b}_{dc}^a \\ \boldsymbol{b}_{lc}^a \end{bmatrix}, \tag{16}$$

where $\boldsymbol{W}_m^a, \boldsymbol{W}_d^a, \boldsymbol{W}_l^a, \boldsymbol{W}_{dc}^a, \boldsymbol{W}_{lc}^a$ and $\boldsymbol{b}_m^a, \boldsymbol{b}_d^a, \boldsymbol{b}_l^a, \boldsymbol{b}_{dc}^a, \boldsymbol{b}_{lc}^a$ are the parameters to be learned. $\alpha_m, \alpha_d, \alpha_l, \alpha_{dc}, \alpha_{lc}$ are the obtained attention weights. Thus, we can calculate the final information importance scores $\boldsymbol{a}$ as

$$\boldsymbol{\alpha} = \text{softmax}([\alpha_m, \alpha_d, \alpha_l, \alpha_{dc}, \alpha_{lc}]), \tag{17}$$

where $\boldsymbol{a} = [\alpha'_m, \alpha'_d, \alpha'_l, \alpha'_{dc}, \alpha'_{lc}]$.

Finally, we obtain the ultimate patient representation vector by summing the heterogeneous information vectors according to the importance scores from Eq. (17), as follows:

$$\boldsymbol{O}_p = \alpha'_m \boldsymbol{h}_{t-1}^m + \alpha'_d \boldsymbol{h}_t^d + \alpha'_l \boldsymbol{h}_t^l + \alpha'_{dc} \boldsymbol{e}_t^{d_c} + \alpha'_{lc} \boldsymbol{e}_t^{l_c}, \tag{18}$$

where $\boldsymbol{O}_p$ is the patient representation vector used to recommend reasonable medications, as described in the next subsection.

3.2.5. Medication recommendation

Doctors make decisions about recommending reasonable medications to patients after comprehensive consideration. Likewise, the learned patient representation $\boldsymbol{O}_p$ is employed in this study to recommend reasonable medications as follows:

$$\hat{\mathbf{y}}_t = \text{softmax}\left(\mathbf{W}_o \cdot \mathbf{O}_p + \mathbf{b}_o\right), \tag{19}$$

where $\hat{\mathbf{y}}_t$ denotes the set of recommended multilabel medications, $\mathbf{W}_o \in \mathbb{R}^{d_m \times r}$ and $\mathbf{b}_o \in \mathbb{R}^{d_m}$ are parameters to be learned.

3.3. Model training

Because the medication recommendation task belongs to the domain of sequential multilabel prediction task, we utilize the binary cross-entropy loss $\mathcal{L}_{ce}$ and multilabel margin loss $\mathcal{L}_{mg}$ as the objective functions. The prediction objective function binary cross-entropy loss $\mathcal{L}_{ce}$ is formulated as follows:

$$\mathcal{L}_{ce} = -\sum_{t=1}^{T} \boldsymbol{y}_t \log \sigma\left(\hat{\boldsymbol{y}}_t\right) + \left(1 - \boldsymbol{y}_t\right) \log\left(1 - \sigma\left(\hat{\boldsymbol{y}}_t\right)\right), \tag{20}$$

The corresponding objective function multilabel margin loss $\mathcal{L}_{mg}$ is:

$$\mathcal{L}_{mg} = \sum_{t}^{T} \sum_{i}^{|\mathcal{C}|} \sum_{j}^{|Y^t|} \frac{\max\left(0, 1 - \left(\hat{y}_t[Y_j^t] - \hat{y}_t[i]\right)\right)}{L}, \tag{21}$$

where $\hat{y}_t[i]$ is the value of the ith coordinate at the tth visit and $\hat{y}_t[Y_j^t]$ denotes the predicted label value indexed by Y_j^t, and jth value in the ground truth label set Y^t at the tth visit for a patient. Thus, we can obtain the binary cross-entropy loss functions $\mathcal{L}_{ce}$

and multilabel margin loss functions $\mathcal{L}_{mg}$. To facilitate the optimization process, we combine the aforementioned loss functions to build a joint loss function $\mathcal{L}$:

$$\mathcal{L} = \eta\mathcal{L}_{ce} + \varepsilon\mathcal{L}_{mg}, \tag{22}$$

where $\eta, \varepsilon \geqslant 0$ are the mixture weights, and $\eta + \varepsilon = 1$. The training algorithm is detailed in Algorithm 1.

Algorithm 1 Model training for MeSIN.

Require: Training set $\boldsymbol{R}$, training epochs N, batch size BS, mixture weights η, ε in Eq. (22) Use uniform distribution to initialize the model parameters $\theta \sim U(-1, 1)$;

for $i = 1$ to $N * |\boldsymbol{R}|$ **do**
  Sample BS patients ($\boldsymbol{P} = \{\mathcal{X}^1, \ldots, \mathcal{X}^{T_n}\}$) from $\boldsymbol{R}$;
  for $t = 1$ to T_i **do**
    /**Laboratory results embedding module**/
    Obtain multichannel time-series embeddings $\{\boldsymbol{h}^l_{1T}, ..., \boldsymbol{h}^l_{qT}\}$ using Eq. (2).
    Obtain enhanced laboratory results embedding $\boldsymbol{e}^{l_c}_t$ using ASM using Eq. (3);
    Compute the visit-level laboratory results embedding $\boldsymbol{h}^l_t$ using Eq. (4);
    /**Diagnoses codes embedding module**/
    Obtain diagnoses codes embeddings $\boldsymbol{E}^d_t = [\boldsymbol{e}^d_1, \ldots, \boldsymbol{e}^d_m] \in \mathbb{R}^{d\times|m}$ using Eq. (5);
    Obtain the enhanced diagnosis code embedding $\boldsymbol{e}^{d_c}_t$ using ASM as Eq. (6);
    Compute the visit-level diagnoses codes embedding $\boldsymbol{h}^d_t$ using InLSTM (Eqs. (7)–(11)).
    /**Medications codes embedding module**/
    Obtain the medications codes embeddings $\boldsymbol{E}^m_z = [\boldsymbol{e}^m_1, \ldots, \boldsymbol{e}^m_n] \in \mathbb{R}^{d\times|n}$ using Eq. (12);
    Obtain the enhanced medications codes embedding $\boldsymbol{e}^{m_c}_{t-1}$ using ASM as Eq. (13);
    Compute the visit-level medications codes embedding $\boldsymbol{h}^m_{t-1}$ using InLSTM (Eqs. (7)–(11)).
    /**Global selective fusion module**/
    Incorporate the multisource embeddings into patient representation $\boldsymbol{O}_p$ using Eqs. (16)–(18).
    Compute recommended medications $\hat{\boldsymbol{y}}^t$ in Eq. (19);
  end for
  Update θ by optimizing the total loss $\mathcal{L}$ in Eq. (22);
end for

4. Experiments and discussion

4.1. Datasets description

As analyzed in Section 1, this study aims to recommend medications for patients based on heterogeneous multilevel EHR data. Hence, we should conduct experiments on a cohort in which patients have at least two visits and their EHRs are complete. Here, we chose a practical publicly available dataset MIMIC-III [44],[1] in which patients stayed within the intensive care units (ICU) at the Beth Israel Deaconess Medical Center and had relatively complete health records with multilevel heterogeneous data. Although MIMIC-III belongs to the ICU, there are certain patients with multiple visits. Hence, we utilized this as our experimental dataset. Similar to [6], we chose the medications prescribed by doctors for each patient within the first 24 h as the medicine set because it is usually a critical period for each patient to obtain rapid and accurate treatment [45]. In addition, the medicine codes form NDC are transformed to ATC Level 3 for integration with MIMIC-III, and since predicting category information not only guarantees the sufficient granularity of all the diagnoses but also improves the training speed and predictive performance [28,31]. To consider the laboratory results in the decision-making process, we followed the feature extraction method used in [46]. Here, the time window of each laboratory indicator was 24 h.Table Table 1 provides more information about the patient cohort from the dataset.

[1] https://mimic.physionet.org.

Table 1
Statistics of the MIMIC-III datasets.

MIMIC III	Quantity
# of patients	4631
# of unique diagnosis	1879
# of unique medication	143
# of unique laboratory indicators	17
avg # of visits	2.55
avg # of diagnoses	10.16
avg # of medications	7.33

4.2. Evaluation metrics

To evaluate the performance, we adopted the Jaccard similarity score (Jaccard), precision recall AUC (PR-AUC), average recall (Recall), and average F1 (F1) as the evaluation metrics. Jaccard is defined as the size of the intersection divided by the size of the union of the predicted set $\hat{Y}^i_t$ and ground truth set Y^i_t. Moreover, for medication recommendation, the number of positive and negative samples are imbalance, the precision–recall curve used to computed the PR-AUC has shown to be a proper metric. Precision is used to measure the correctness of predicted medicines, while recall is used to measure the completeness of the predicted medicines. Our MeSIN model cannot completely replace doctors and only assist doctors in prescribing medicines for patients. Thus, the critical task is to help doctors screen possible medicines as much as possible. In addition, F1 is often used as a comprehensive evaluation metric for multilabel classification tasks.

$$\text{Jaccard} = \frac{1}{\sum_i^N \sum_t^{T_i} 1} \sum_i^N \sum_t^{T_i} \frac{\left|Y^i_t \cap \hat{Y}^i_t\right|}{\left|Y^i_t \cup \hat{Y}^i_t\right|}, \tag{23}$$

where N denotes the number of patients in the test set, and T_i is the number of visits for the ith patient. Given

$$\text{Recall} = \frac{1}{\sum_i^N \sum_t^{T_i} 1} \sum_i^N \sum_t^{T_i} \frac{\left|Y^i_t \cap \hat{Y}^i_t\right|}{\left|Y^i_t\right|}, \tag{24}$$

$$\text{Precision} = \frac{1}{\sum_i^N \sum_t^{T_i} 1} \sum_i^N \sum_t^{T_i} \frac{\left|Y^i_t \cap \hat{Y}^i_t\right|}{\left|\hat{Y}^i_t\right|}, \tag{25}$$

the comprehensive evaluation metric F1 can be calculated as follows:

$$\text{Jaccard} = \frac{1}{\sum_i^N \sum_t^{T_i} 1} \sum_i^N \sum_t^{T_i} \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{26}$$

4.3. Benchmark methods

To evaluate the effectiveness of the proposed model, it was compared with the following baseline methods or feasible alternatives:

- *Nearest*. To predict treatment medicines for a patient p_i, Nearest was proposed to choose the treatment medications prescribed for patient p_j, who has the most similar historical laboratory indicators and medications as p_i.
- *LR [47]*. It is a logistic regression with L1/L2 regularization. We sum the multihot vector of each visit together and apply the binary relevance technique [47] to handle multilabel output.
- *NBN [48]*. This method mainly utilizes prior knowledge and employs statistical methods to recommend the corresponding medications for patients.
- *Retain [30]*. RETAIN is an interpretable model with a two-level reverse time attention mechanism to predict diagnoses, which can detect significant past visits and associated clinical variables. It can be used for similar sequential prediction tasks, such as the prediction of treatment medicines.
- *DELSTM [11]*. This model utilizes an additional input sequence as the input of a decomposed gate to control the memory cell state, which indirectly interacts with the primary input sequence.
- *PCLSTM [11]*. This structure takes all the heterogeneous sequences as inputs. In other words, multiple sequences interact with each other via neuron interactions by concatenating both hidden states.
- *RAHM [12]*. It builds a relation-augmented hierarchical multi-task learning framework for learning multilevel relation-aware patient representation for medication prediction.
- *LEAP [7]*. Leap formulates the medicine prediction problem as a multi-instance multilabel learning problem, which mainly uses a recurrent neural network (RNN) to recommend medicines.
- *DMNC [9]*. DMNC uses a memory-augmented neural network to model the interaction of two asynchronous sequences for treatment prediction tasks [9].
- *GAMENet [6]*. It employs a dynamic memory network to save encoded historical medication information, and further utilizes a query representation formed by encoding sequential diagnosis and procedure codes to retrieve medications from the memory bank.
- *AMANet [10]*. AMANet leverages self-attention and inter-attention to capture intra-view and inter-view interactions. Then, it concatenates the information from history attention and dynamic external memory to predict the medications.

4.4. Experimental settings

We randomly split the patients in the MIMIC-III dataset into training, validation, and test sets with 2/3 : 1/6 : 1/6 ratios. Random splitting and training processes were performed five times. Table 2 lists the results as averages across the five runs obtained for all the compared models in terms of the four evaluation metrics described in Section 4.2. In particular, the embedding size and hidden layer dimension for LSTM and GRU are all set to 128 and 128, respectively. In this case, the dimension of rest parameters could be calculated according to the equations in the paper. The dropout rate was set to 0.4, the batch size was set to 10, and the mixture weights of the objective function were set as $\eta = 0.99, \varepsilon = 0.01$. The values of the attention sparsity control parameters $(\gamma_l, \gamma_d, \gamma_m)$ in the ASM are set to 1.5, 1.5, and 1.3, respectively. Training was performed using Adam [49] at a learning rate of 2e–4, and we report the model performance in the test set within 40 epochs. All methods were trained on a Ubuntu 16.04 with 64 GB memory and an Nvidia TAITAN XP GPU using the Pytorch 1.0 framework.

Table 2
Performance comparison of benchmark methods on MIMIC-III dataset.

Methods	Jaccard	PR-AUC	Recall	F1
Nearest	0.2019	0.2227	0.3099	0.2899
LR [47]	0.3401	0.5549	0.4549	0.4901
NBN [48]	0.3341	0.5479	0.5081	0.4839
Retain [30]	0.3267	0.5364	0.4841	0.4905
DELSTM [11]	0.3357	0.5371	0.5144	0.5005
PCLSTM [11]	0.3301	0.5109	0.4952	0.4940
RAHM [12]	0.3558	0.5393	0.5357	0.5122
LEAP [7]	0.3111	0.4212	0.4541	0.4627
DMNC [9]	0.3272	0.4476	0.5136	0.5021
GAMENet [6]	0.3452	0.4828	0.5246	0.5115
AMANet [10]	0.3641	0.5595	0.5547	0.5381
MeSIN	**0.3975**	**0.5684**	**0.5934**	**0.5670**

4.5. Performance comparison

As demonstrated in Table 2, the benchmark models used in health informatics are divided into three categories: shallow methods, including Nearest, LR, and NBN; predictive models, including Retain, DELSTM, PCLSTM, and RAHM; recommendation models, including LEAP, DMNC, GAMENet, and AMANet. The table indicates that the proposed MeSIN achieves superior performance over the listed benchmark models. Through a detailed comparison with the benchmark models, several interesting observations can be made.

First, for the shallow methods, Nearest, LR, and NBN achieved approximately 5.74%, 1.35%, 8.53%, and 7.69% lower scores on the medication recommendation task with respect to Jaccard, PR-AUC, Recall, and F1 score, respectively, than MeSIN. On the one hand, this type of method does not consider the temporality and heterogeneity of EHR data and overlooks the relationships between multiple sequences. On the other hand, Nearest achieves the worst performance, which indicates that most patients possess distinct disease conditions within continuous hospital admissions.

Second, as for predictive models in health informatics, Retain, DELSTM, PCLSTM, and RAHM also achieved poorer performance on medication recommendation tasks than MeSIN. We think that the main reason might be that they cannot consider the current medical records, including laboratory indicators and diagnosed disease status, and only take the historical records into account. In addition, Retain is a two-level attention-based model that can capture temporal correlations and identify influential past visits. However, it cannot consider all heterogeneous sequences and can simply concatenate the heterogeneous embeddings into one embedding, which would confuse the embeddings obtained from different hierarchies of EHR data. DELSTM and PCLSTM mainly focus on the sequence interaction in the temporal sequence learning process while overlooking the inherent hierarchy structure of EHR data. In MeSIN, a multilevel learning framework is incorporated to extract useful information from such an inherent structure. RAHM partially employs the hierarchical nature of EHR data to achieve better performance via a multitask learning framework. However, it still employs single sequence learning methods to integrate historical information, which might cause confusion in different historical medical sequences.

Third, our proposed MeSIN outperforms all state-of-the-art methods used for medication recommendation, such as LEAP, DMNC, GAMENet, and AMANet by 8.64%, 14.72%, 13.93% and 10.43%, and at least 3.34%, 0.92%, 3.87%, and 2.89% with respect to the Jaccard, PR-AUC, recall, and F1 score. In practice, the medication recommendation problem within an admission might not be a pure recommendation process, and it also refers to the diverse interaction process. Their poor performance could

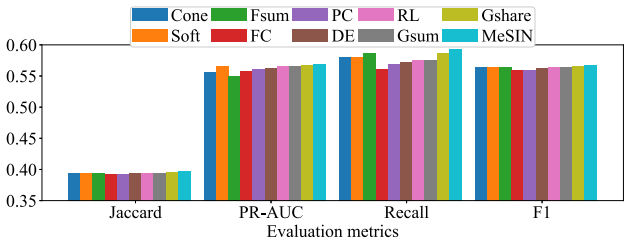


Fig. 4. Performance comparison of MeSIN with alternative components.

be attributed to their poor ability to capture such complicated interactions. In particular, LEAP cannot capture the inherent multiple relations among heterogeneous sequences. DMNC realizes the interactions of two sequences through attention-based DNC blocks, but neglects the utilization of medications in the history visits. Similarly, AMANet also does not consider historical medications prescribed for patients, but it achieves relatively better performance through multiple attention networks for capturing the inter- and intra-correlations of heterogeneous sequences. However, AMANet neglects the captured evolution information, such as disease progression, through a sequence learning network, which is still important information in the decision-making process.

Therefore, the critical reasons that MeSIN achieves the best performance compared with all benchmark models can be summarized as follows: (1) The multilevel learning framework can help capture the inherent causal relations of adjacent hierarchies. (2) The incorporated attentional selective module (ASM) in the framework realizes effective embedding selection, which further facilitates the subsequent hierarchical interaction process. (3) The novel InLSTM further reinforces the sequential interactions, which can further optimize the temporal sequence learning process by incorporating more useful calibrated information.

4.6. Analysis on the components of MeSIN

To verify the effectiveness of the critical components of MeSIN for medication prediction, we first utilized feasible alternatives to replace them for validating their superiority. Then, we conducted an ablation study to further explore the necessity of the proposed model components in hierarchically connected multilevel modules.

4.6.1. Feasible alternatives of model components

Fig. 4 shows the prediction performance of MeSIN with alternative components. They are **Cone** in which *multichannel time-series embedding layer* is replaced with normal *one-channel time-series embedding layer* that directly takes q-dimensional vector as input, **Soft** in which the attention weights computation method *Entmax* in ASM is replaced with *Softmax*, **Fsum** in which module *ASM* is replaced with simple *sum* operation, **Gsum** in which module *GSFM* is also replaced with simple *sum* operation, **Gshare** in which the self-attention mechanism in *GSFM* is replaced with an general attention mechanism, and **FC**, **PC**, **DE**, **RL** in which *InLSTM* is respectively replaced with *FC-LSTM* [11], *PC-LSTM* [11], *DE-LSTM* [11] and *R-LSTM* [12].

As shown in Fig. 4, we can observe that our MeSIN model still outperforms all the aforementioned variants with feasible alternative components in MeSIN. The reasons can be summarized as follows. For **Cone**, the alternative component of the one-channel time-series embedding layer makes MeSIN unable to capture the distinct progression of laboratory indicators. In **Soft**, *Softmax* generates dense attention alignments that are wasteful and cannot focus more on important feature embeddings. Similarly, **Fsum** chooses to simply sum all the feature embeddings without any selection, which can exert a more negative influence on the results because of the noisy feature embeddings. The same is true for the **Gsum**. In addition, **Gshare** employs a shared parameter matrix to compute attention weights, which overlooks the heterogeneity of learned information embeddings. For MeSINs with alternative sequential interaction methods, *PC-LSTM* and *FC-LSTM* cannot capture the complicated sequence interaction process simply through a simple concatenation operation of neurons of the LSTM cell. Although *DE-LSTM* and *R-LSTM* consider the impact of auxiliary input on the memory cell state of the primary input sequence learning network through specially designed interactive network structures, they all overlook the impact on the current input cell state of the sequence learning network. However, the proposed InLSTM network in MeSIN simultaneously considers sequential interactions from the above two aspects.

4.6.2. Ablation study on model components

This ablation study was conducted to verify the effectiveness of different MeSIN components on the overall performance. To determine whether the incorporated components improved the performance, we added them one by one from scratch and verified their performance using all evaluation metrics including Jaccard, PRAUC, recall, and F1 score. Table 3 presents the recommendation results of distinct MeSIN variants on the MIMIC-III dataset. One of the basic baseline models, Vanilla, the medical codes are added together as the enhanced embeddings in every module. In addition, the standard LSTM networks are employed as the temporal sequence learning networks in three distinct modules, and we employ the concatenation-based fusion method to replace the proposed global selective fusion module. However, Vanilla still achieves relatively better performance compared with benchmark models, which can be attributed to the incorporation of multisource data and the integration of current laboratory results and disease diagnosis using the concatenation-based fusion method.

Attentional selective module (ASM). As explained in Section 3.2, ASM is introduced to automatically select useful information and filter out noise information as much as possible by assigning corresponding attention weights to embeddings according to their respective importance. The following variants were tested to evaluate the contribution of ASMs from different modules to the overall performance of MeSIN:

- ASM_L. In this variant, we incorporate an attentional selective module for the embedding selection of laboratory results. The overall performance is slightly improved by 0.17% on Jaccard in this case compared with the Vanilla model. This indicates that the introduced ASM module can help focus on the useful laboratory results embeddings by controlling the value of γ_l, by which we can obtain relatively better enhanced embedding as the input of the temporal sequence learning network.
- ASM_{LD}. Similar to ASM_L, in this variant, we introduce an attentional selective module to replace the addition operation for diagnosis code embedding selection. In this manner, the diagnoses code embeddings that are irrelevant to the recommendation task are discarded by sparse attention under the value control of γ_d. As a result, the performance of ASM_{LD} is improved by 0.16% on Jaccard, which indicates the importance of the ASM in MeSIN in selecting useful information from numerous medical code embeddings.
- ASM_{LDM}. In this variant, we further incorporate the third ASM module into the prescribed medication embedding module to select the most relevant historical medication code embeddings to build the patient representation. As a result, ASM_{LDM} provides a relatively better improvement compared with ASM_L and

Table 3
Performance comparison of the variants of MeSIN on MIMIC-III dataset.

Model	ASM			InLSTM		GSFM	Recommendation performance			
	Lab	Diag	Med	Diag	Med	Fusion	Jaccard	PR-AUC	Recall	F1
Vanilla	✗	✗	✗	✗	✗	✗	0.3832	0.5579	0.5451	0.5482
ASM_L	✓	✗	✗	✗	✗	✗	0.3849	0.5592	0.5522	0.5501
ASM_{LD}	✓	✓	✗	✗	✗	✗	0.3865	0.5595	0.5549	0.5546
ASM_{LDM}	✓	✓	✓	✗	✗	✗	0.3887	0.5592	0.5684	0.5583
ASM _$InLSTM_D$	✓	✓	✓	✓	✗	✗	0.3916	0.5643	0.5673	0.5619
ASM _$InLSTM_{DM}$	✓	✓	✓	✓	✓	✗	0.3935	0.5651	0.5768	0.5639
$InLSTM_{DM}$	✗	✗	✗	✓	✓	✗	0.3921	0.5641	0.5643	0.5615
$GSFM_{alone}$	✗	✗	✗	✗	✗	✓	0.3869	0.5590	0.5561	0.5509
MeSIN	✓	✓	✓	✓	✓	✓	**0.3975**	**0.5684**	**0.5934**	**0.5670**

ASM_{LD}. We believe that this can be attributed to the fact that the medication embedding module has direct relevance to the medication recommendation task. In the end, the incorporation of the above three attentional selective modules brings about 0.55% on Jaccard, 0.13% on PR-AUC, 2.33% on Recall, and 1.01% on F1 in total compared with Vanilla. However, ASM makes MeSIN more interpretable by focusing on the very important features.

Interactive Long-Short Term Memory network (InLSTM). InLSTM was developed to reinforce the interaction process of heterogeneous sequences, which is beneficial for capturing the correlations of sequences. The following variants were tested to evaluate the contribution of InLSTM to the overall performance of MeSIN:

- ASM _$InLSTM_D$. In this variant, we incorporate a novel InLSTM to replace the standard LSTM in ASM_{LDM} in the diagnosis code embedding module to enhance the interaction process of disease progression and changing laboratory results. It achieves by 0.29% on Jaccard compared with ASM_{LDM}, which verifies the importance of considering the correlations of sequences such as between laboratory results and diagnosed diseases into the temporal sequence learning process.
- ASM _$InLSTM_{DM}$. Here, the interactive LSTM is further introduced to the top hierarchy, a prescribed medication embedding module for facilitating the medication code sequential learning process. In this sequence learning network, the diagnosed diseases are utilized to enhance the interaction process with prescribed medications to provide complimentary useful information. Thus, the performance of ASM _$InLSTM_{DM}$ outperforms fifth variant ASM _$InLSTM_D$ by 1.9% on Jaccard, further indicating the superiority of InLSTM in MeSIN over the standard LSTM in ASM_{LDM}. $InLSTM_{DM}$. In this variant, we removed all the introduced components except InLSTM in MeSIN. We can observe that the performance of $InLSTM_{DM}$ is better than ASM_{LDM}, which indicates that the positive impact of InLSTM is stronger than that of ASM. This also shows that the negative impact on the prediction results from the noisy information within the input after data reprocessing is not as serious.

Global selective fusion module (GSFM). After step **I**, the hierarchically interactive temporal sequence learning procedure, the obtained multisourced embeddings are integrated together via proposed global selective fusion module (GSFM) for obtaining the patient representation. In this manner, MeSIN can automatically learn the contribution scores of distinct embeddings to the medication recommendation task. As a result, it improves by 0.4% on Jaccard compared with the sixth variant ASM _$InLSTM_{DM}$. This also indicates the advantage of the GSFM over the summation-based method used in the above six variant models. In addition, variant $GSFM_{alone}$ which only possesses component GSFM achieves better performance than Vanilla, which further testifies the positive influence of GSFM.Finally, owing to the utilization of the summation-based fusion method and the fusion of multisourced heterogeneous EHR data, Vanilla still achieved a relatively better performance than the benchmark methods.

Table 4
Contribution of different data sources to MeSIN performance.

Methods	Jaccard	PR-AUC	Recall	F1
NoLab	0.3861	0.5612	0.5526	0.5567
NoDiag	0.3552	0.5548	0.5158	0.5236
NoMed	0.3859	0.5587	0.5514	0.5537
AllData	**0.3975**	**0.5684**	**0.5934**	**0.5670**

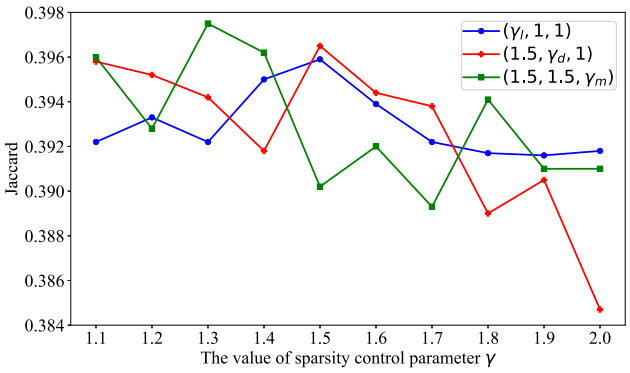
4.7. Influence of heterogeneous data

According to the proposed method, multilevel EHR data must be input to MeSIN to obtain the final patient representation. Although each of them plays a paramount role in the clinical decision-making scenario, here, we would build the following MeSIN variants to evaluate the impact of different heterogeneous data on medication recommendation results (Table 4). In *NoLab*, the laboratory results embedding module in MeSIN is removed, and the introduced $InLSTM_d$ in the diagnosis code embedding module needs to be replaced by the standard LSTM network. In this case, the patient's detailed health status was unknown. In NoDiag, the diagnosis code embedding module is removed from MeSIN and only retains the remaining two modules. Under such circumstances, the learned patient representation will lose key disease progression information. In NoMed, the medication code embedding module is removed from the MeSIN. In this manner, the learned patient representation will lose historical medication information.

Table 4 indicates that the performance of all variants drops owing to some apparent reasons. First, most medications are prescribed for the diagnosis of diseases. Therefore, the performance of *NoDiag* drops dramatically, which validates the crucial role of diagnosed diseases and disease progression in medication recommendation tasks. Second, although historical medications in the ICU are not very valuable for most patients, they are still important information in understanding patients' history diseases, which can help understand some detailed information such as allergic conditions. Thus, the performance of *NoMed* also drops significantly in the medication recommendation task. Third, the performance of *NoLab* also drops significantly in this case but slightly compared with *NoDiag*. The main reason is that the current patient health status, including diagnosed diseases and key laboratory indicator results, is still a paramount indicator indicating the patient's health status. Thus, although the importance of each hierarchy data within EHRs is diverse. from each other, all of them play important roles in medication recommendation task.

4.8. Analysis on the hyper-parameters

To explore the influence of hyper-parameters, including the sparsity control parameter γ, the embedding size of the medical



Performance comparison of MeSIN with different values for sparsity control parameters γ in ASM.

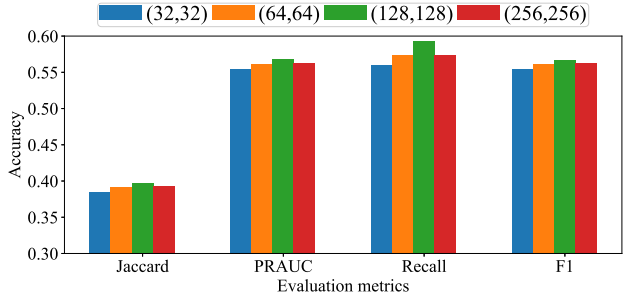


Fig. 5. Performance comparison of MeSIN with different sparsity control parameter γ in ASM (top),embedding size and dimension of hidden state (bottom).

codes, and the hidden layer dimension, we conducted experiments with different values.

For the sparsity control parameters γ in ASM module, Fig. 5 shows the recommendation performance (concerning Jaccard) of MeSIN with different values. In the figure, $(1.5, 1.5, \gamma_m)$ denotes a special case in which γ_l and γ_d are set to 1.5, and γ_m is set to different values. The other two cases were similar. We can observe that the recommendation performances are different with distinct values for the sparsity control parameter γ in the respective ASM modules. This phenomenon indirectly testifies that the impact of heterogeneous information on recommendation results is different. In addition, with the increase in the value of γ, the recommendation performance is almost all declining, which indicates that some helpful information is also deleted. This also demonstrates that diagnosis information is the most important information for medication recommendation. Finally, the experiments validate that the best values of γ in different ASM modules are 1.5, 1.5, 1.3. Two other important hyper-parameters are the embedding size and hidden layer dimension. As shown in Fig. 5, we find that the best results occur when the dimension values are set to 128. When the dimensions of the hidden state and other related parameters are too small, the mapping relations between the inputs and outputs are not fully built, which results in a relatively poor recommendation performance. In addition, the change in recommendation performance with different parameter values is not too large, which indicates that they are not sensitive to the results. Therefore, to maintain the balance between recommendation accuracy and computation cost, the dimension values were all set to 128. When encountering complicated conditions, the dimension values can be set to a larger value for better server performance.

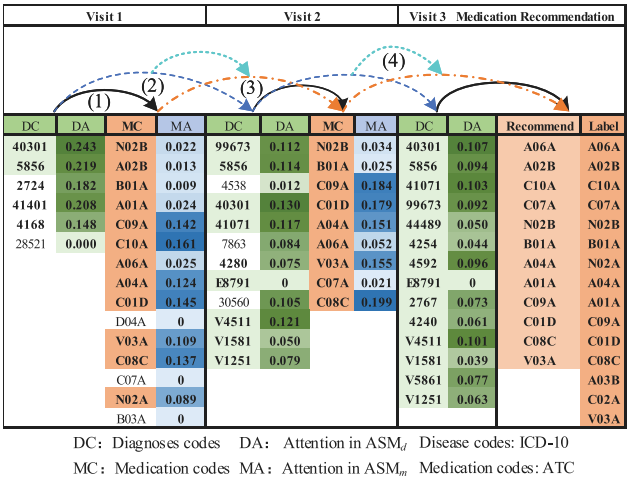


Fig. 6. Visualization of attentive process between the second and third hierarchy within our multilevel learning framework.

4.9. Case analysis

As discussed above, our newly developed MeSIN outperforms all benchmark models on medication recommendations for patients. Thus, in this section, we will take the selective module as an example to further analyze how the constituent components in MeSIN help make a comprehensive decision on treatment medication prediction. In particular, we first choose a patient from the test dataset based on the consideration of illustrating the complicated decision-making process via harder cases: there are multiple disease complications and treatment medications among visits. As shown in Fig. 6, when a patient went to the hospital for a third time, after a series of laboratory examinations and diagnoses, and further concatenating with historical medical records (generated in the first and second observations), our proposed model MeSIN can preliminarily provide the treatment medicines in advance on the computer as a reference for doctors when making comprehensive decisions. In this manner, although the prediction outcome might not be all right, the prediction results can also assist doctors in improving decision-making efficiency, particularly for junior or inexperienced doctors; in other words, it might be useful for alleviating the lack of medical resources.

Subsequently, we will perform a detailed case analysis to firstly explore the crucially attentive process across the hierarchical shown in Fig. 6, then visualize the difference of softmax and entmax shown in Fig. 7, and finally investigate the importance of multisource embeddings shown in Fig. 8.

The attentive process. To clearly interpret the attentive process, as shown in Fig. 6, we only consider the relations between the second and third hierarchies (diagnosis code embedding module and prescribed medication embedding module) within our multilevel learning framework. In addition, the quantitative values in columns DA and MA denote the attention weights calculated using Eqs. (6) and (13). As shown in Fig. 6, the attentive process can be categorized into four distinct but correlated processes. In this case, we made four interesting observations. First, in the attentive process (1), the learned visit-level diagnosis code embedding will be input into the medication code embedding module for interacting with the medication codes embedded within historical visits. Thus, we can observe that strong causal relations exist between the *DC* column (diagnoses codes) and *MC* (medication codes) within each visit. In this manner, the diagnosis codes that correspond to the medication codes existing in the recommendation label column will be assigned more attention weights within each visit. Second, owing to the temporal

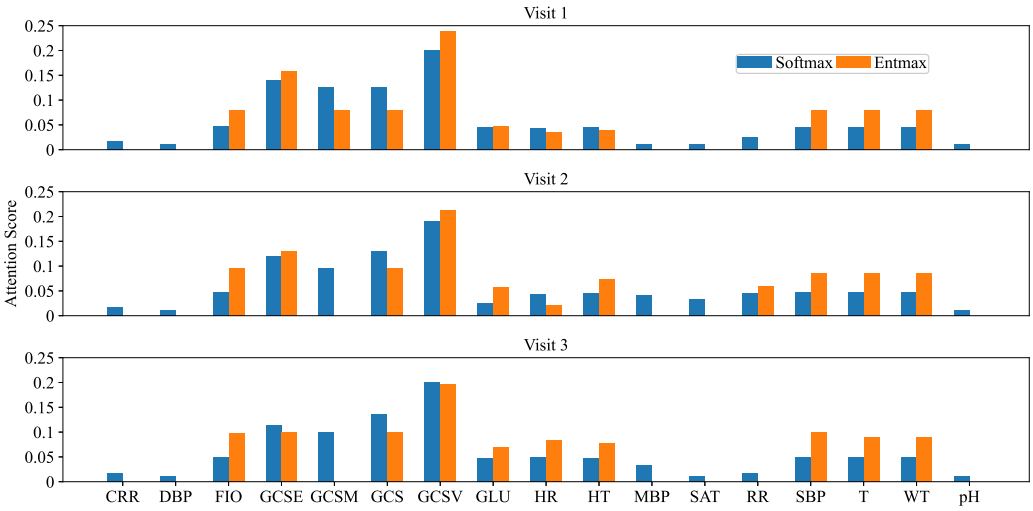


Fig. 7. Comparison of attention weights computed by Entmax and Softmax in laboratory results embeddings selection module.

dependencies of EHR data, the recommended medications in the *Recommend* column not only depend on the diagnosed diseases in the *DC* column in the third visit, but also relies on the historical prescribed medications and disease progression. As for this, as shown in the attentive process (2), the medication code embeddings in historical visits but existing in the *label* column will be assigned more attention weights. Similarly, in the attentive process (3), as for the inherent causal relations between diseases and medications, the corresponding diagnoses code embeddings will also be assigned more attention weights. Finally, as shown in the attentive process (4), to capture the temporal dependency of EHR data, the medication code sequence learning process will be influenced by the diagnosis code sequence learning process under the help of the proposed InLSTM in MeSIN.

The calculation methods of attention weights: *Softmax* and *Entmax*. In MeSIN, *Entmax* has been incorporated into the attentional selective module (ASM) to make more intelligent selections, assisting in filtering out noisy information and paying more focus on the important feature embeddings. In Fig. 7, we provide a hot map that demonstrates the difference in attention weights computed by *Entmax* [42] and *Softmax* in the laboratory results embedding selection module. As shown in the figure, we observe that the calculation method *Entmax* can generate sparse attention weights within each visit; in other words, it can make the attention scores of some unimportant indicator results embeddings to zero, such as CRR, DBP, MBP, OS, RR, and PH. In this manner, the MeSIN can intelligently make selections about which feature embeddings are more important to be focused on and which are unnecessary to be paid so much attention when making decisions in the clinical decision-making process. Therefore, such attention weight computation methods in ASMs can help MeSIN increase focus on the crucial feature embeddings and make the model more interpretable.

The importance of multisource embeddings. As mentioned in Section 3.2.4, the global selective fusion module, to fuse the obtained five heterogeneous embeddings, we introduce a global selective fusion module, which can integrate them into patient representation according to their respective importance scores and minimize the adverse effects caused by noisy information. In Fig. 8, we can observe that $\alpha'_d > \alpha'_{lc} > \alpha'_m > \alpha'_{dc} > \alpha'_l$ (see details in Eqs. (16)–(18)), which indicates the importance ranking of multisource embeddings. This phenomenon further testifies that diagnosed diseases, particularly disease progression, with historical disease information being the most important information for the medication recommendation task, which has

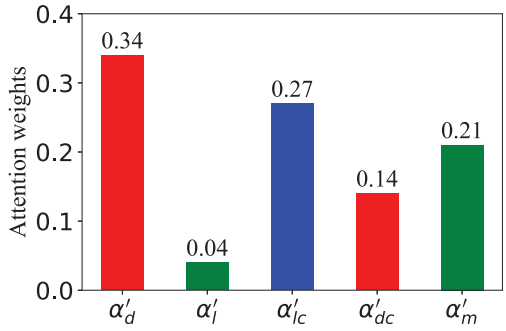


Fig. 8. Attention weights distribution in global selective module.

been proved in the ablation study shown in Table 4. The current laboratory results are the second critical factor when making decisions regarding the recommended medications. In addition, historical prescribed medications were also taken into account. Finally, the historical laboratory results might not be very important in the intensive care unit (ICU). However, because different patients might have different disease statuses, the learned attention weights are also dynamically changing, which makes the computed relevance score distribution also diverse. For example, historical medications might be more important than diagnosed diseases when the diagnosis is an adverse drug reaction. Through the above analysis, we can see that MeSIN can provide insightful and interpretable recommendation results.

5. Conclusion and further study

In this paper, a multilevel selective and interactive network, MeSIN, is proposed for medication recommendation task with clinical EHR data. In the framework, the inherent causal relations and temporal dependencies inherent in the hierarchical multilevel structure of EHR data are effectively captured via the proposed multilevel learning framework and a novel interactive learning network (InLSTM). To enhance the interaction process in InLTM, the enhanced medical code embeds at each timestamp of each hierarchy are obtained through an improved attentional selective module (ASM) by reducing the inevitable noisy information from the input EHR data. Finally, the framework leverages a self attention-based globally selective fusion module to integrate the learned heterogeneous information embeddings into a

comprehensive patient representation for recommending medications. The proposed method was evaluated on a practical and public clinical dataset. The experimental results show that our model achieves the best recommendation performance against 11 baselines in terms of Jaccard, PR-AUC, Recall, and F1 score.

In the future, we plan to adapt the proposed approach for more healthcare prediction tasks based on sequential data and explore its usage in domains other than healthcare. For instance, in the field of industrial automation, the time-series data generated by many sensors in the production process of a component can be used for anomaly event detection with the help of MeSIN. To forecast power consumption, if there are multilevel data including the temporal temperature data, the temporal sales data of air conditioners and other refrigeration equipment, and the temporal historical power consumption data, MeSIN might be used to capture the correlations between adjacent sequences to forecast future power consumption.

CRediT authorship contribution statement

Yang An: Conceptualization, Methodology, Software, Validation, Writing – original draft, Formal analysis, Data Curation, Experiments, Writing – review & editing. **Liang Zhang:** Supervision, Writing – review & editing. **Mao You:** Supervision. **Xueqing Tian:** Supervision. **Bo Jin:** Supervision, Funding acquisition. **Xiaopeng Wei:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.knosys.2021.107534.

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